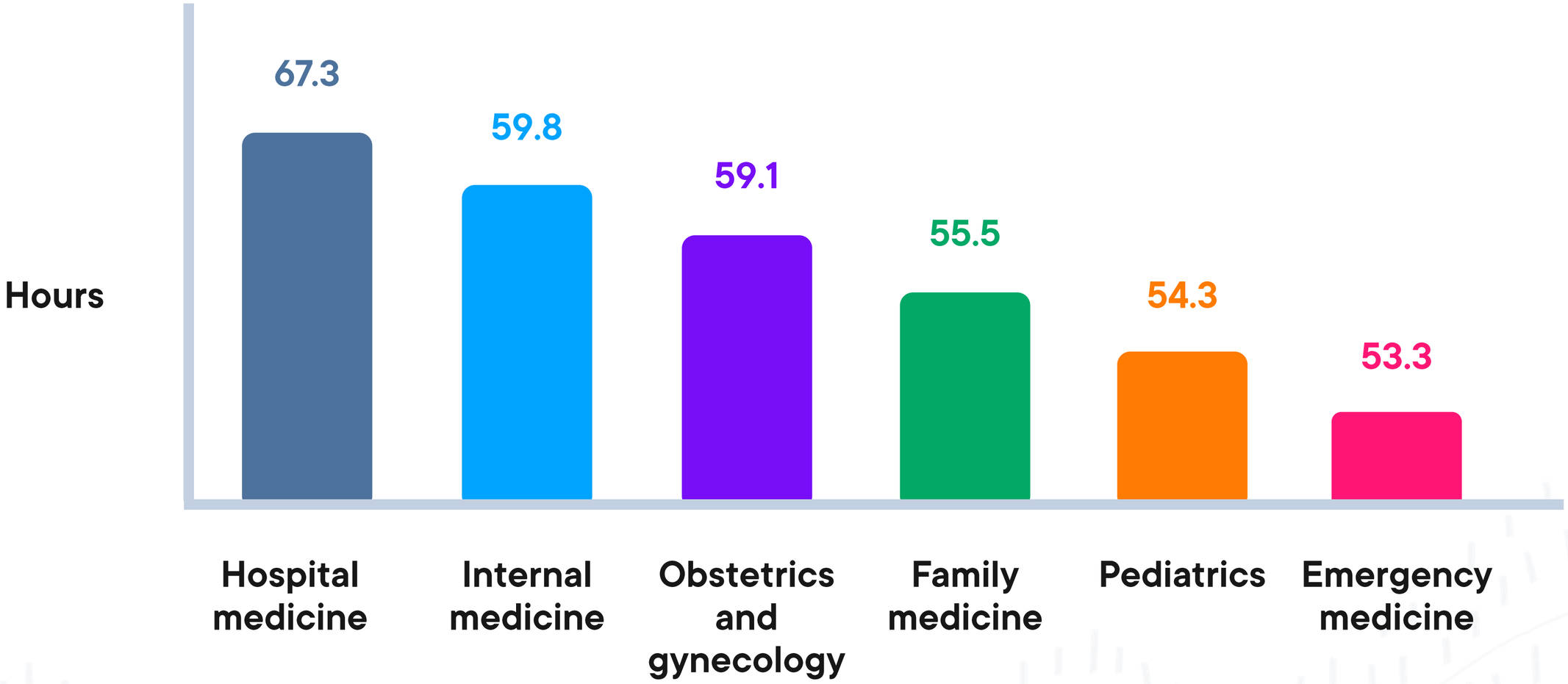
Agentic AI Integrated with Healthcare Workflow

Lyron Andrews, Ed.D.
ISO/IEC 42001 AI Management System Lead Implementor

@drlyronhandrews | www.profabula.com

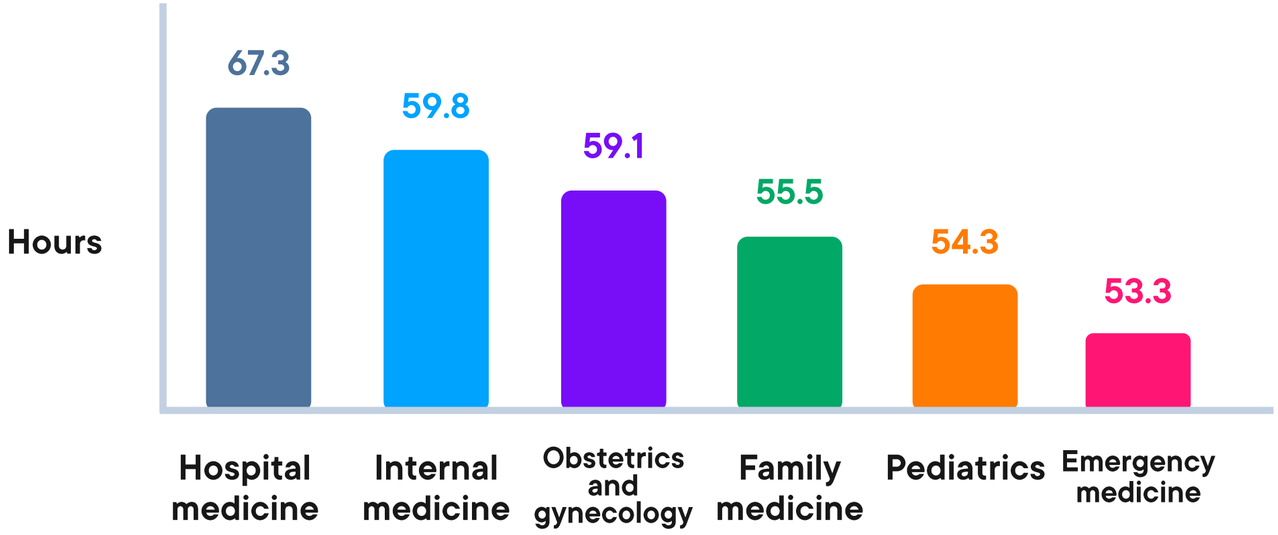
Identifying Opportunities for Operational Efficiency

Physician Administrative Overload

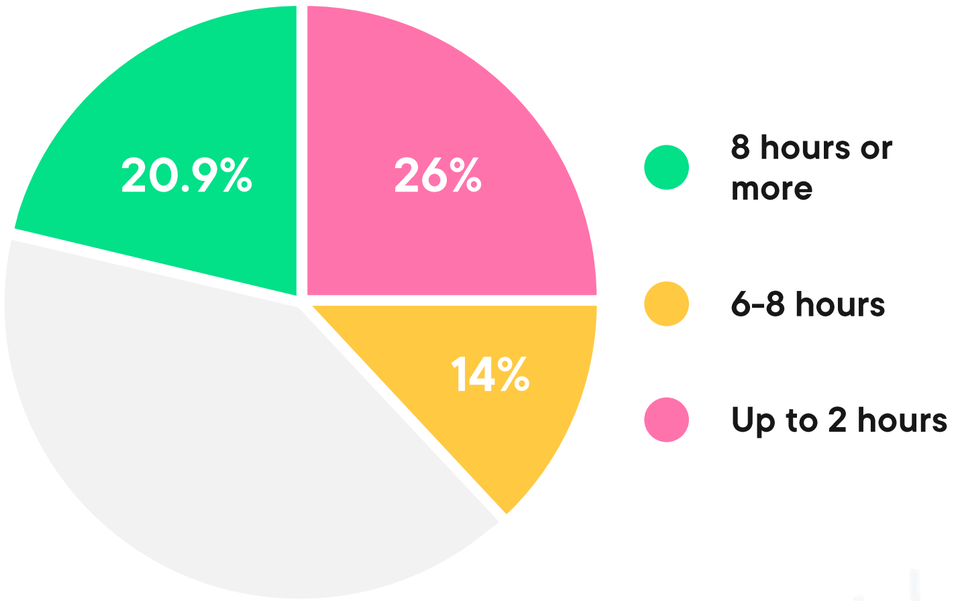


Physician Hours Spent by Practice

Physician Administrative Overload

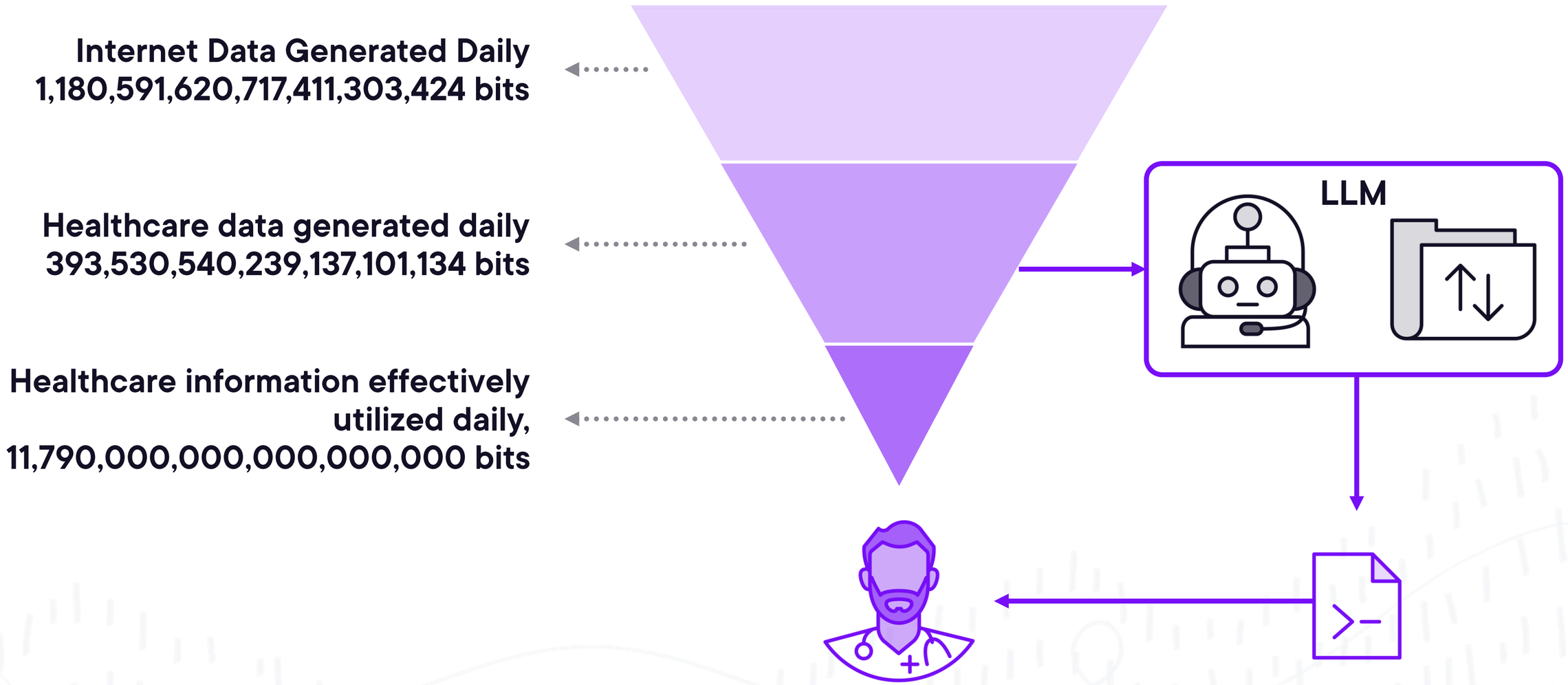


Physician Hours Spent by Practice



EHR Work Outside Normal Hours

Physician Information Overload



Agentic AI Healthcare Automation Opportunities

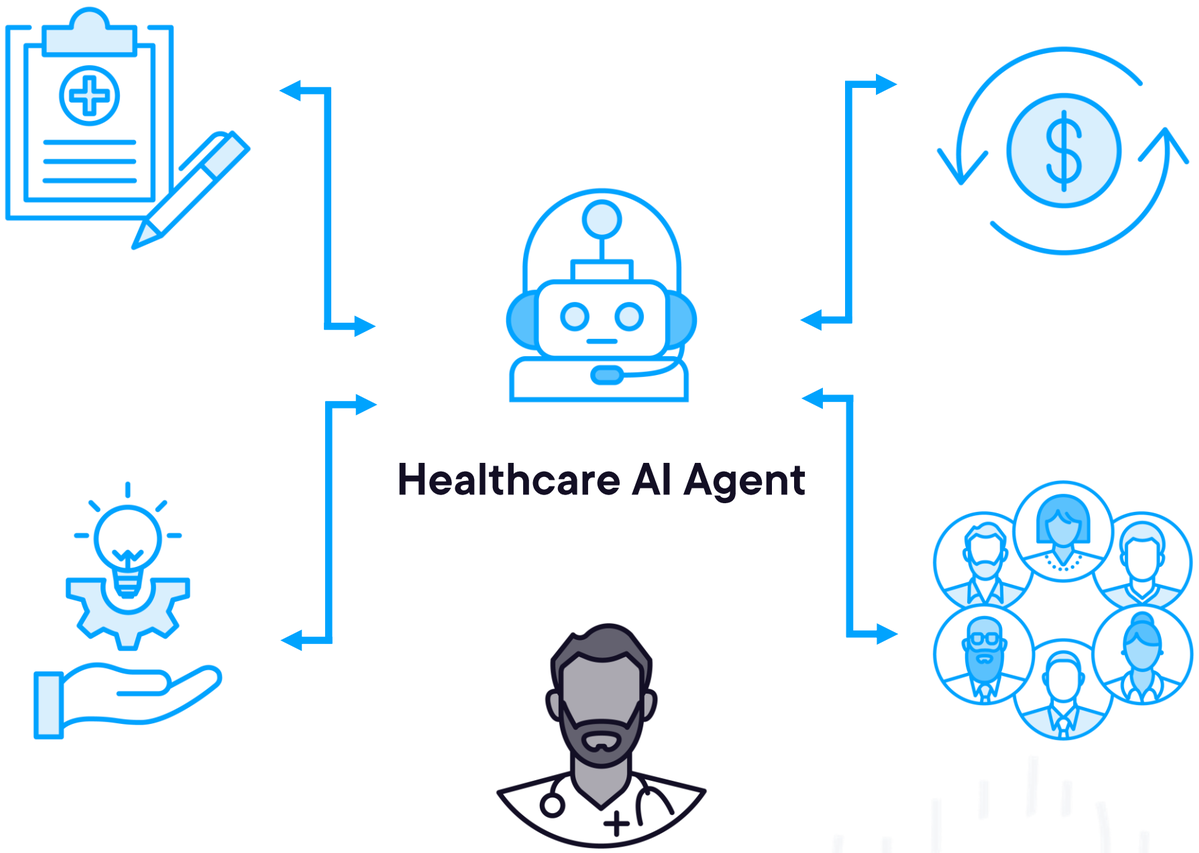
AI and Automation Opportunities

Patient Flow and Resource Management

Streamlining:
Patient flow
Bed management
Resource allocation
Capacity management

Technology Adoption

EHR
Telemedicine
Automation
Data analytics



Revenue Cycle Management (RCM)

Optimizing RCM
Claims processing
Contract configuration

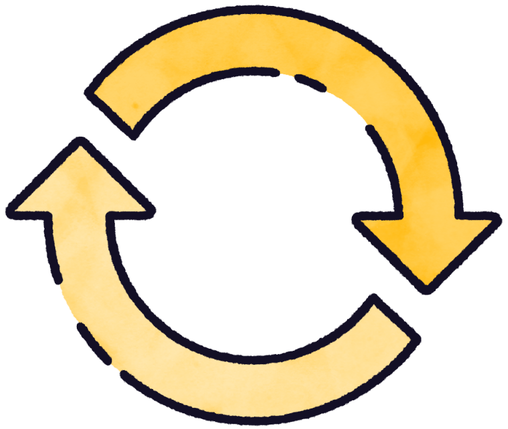
Personnel Support

Staff training and morale
Communication and collaboration
Process standardization
Continuous improvement

Best Practices for Integrating Agentic AI in Healthcare Operations

Seismic changes are transpiring in the US Federal government and can affect the existence of agencies referenced in this clip.

FDA - Digital Health Center of Excellence



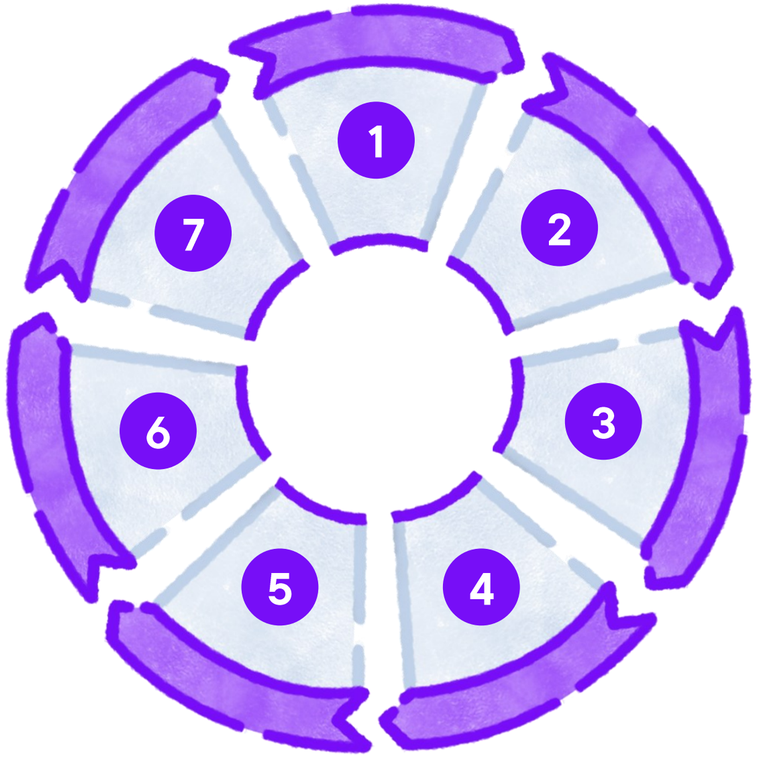
AI Lifecycle

A framework for cyclical phases of AI use and production

AI Medical Products

Advancing products and patient care

AI Lifecycle Applied to Healthcare



AI lifecycle

1. Planning and Design
2. Data Collection and Management
3. Model Building and Tuning
4. Verification and Validation
5. Model Deployment
6. Operation and Monitoring
7. Real-world Performance Evaluations

Risk management and cybersecurity apply throughout the lifecycle

Artificial Intelligence and Medical Products Guidance

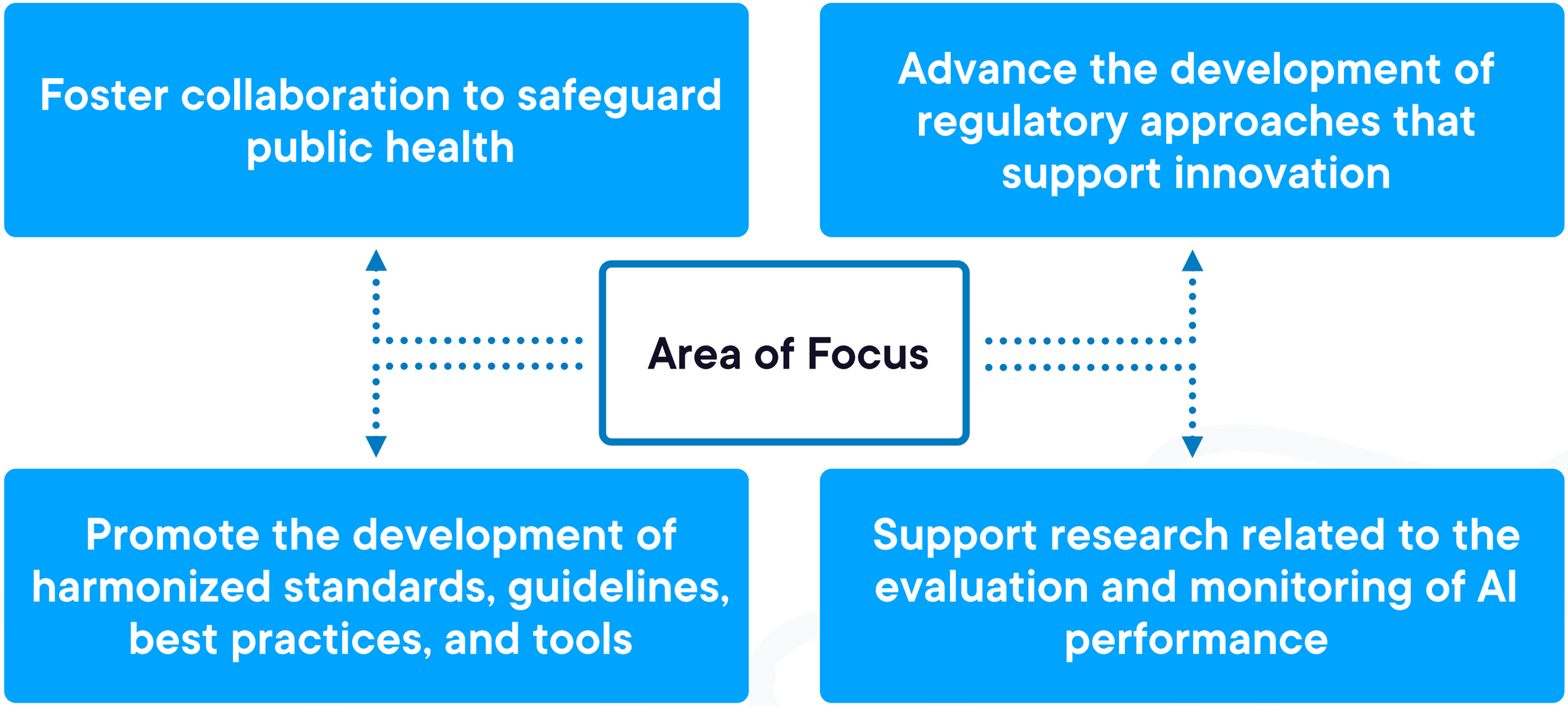
Center for Biologics Evaluation and Research (CBER)

Center for Drug Evaluation and Research (CDER)

Center for Devices and Radiological Health (CDRH)

Office of Combination Products (OCP)

AI Lifecycle Focus Areas



Supporting AI-related Patient Privacy

HIPAA

“The Health Insurance Portability and Accountability Act (HIPAA) of 1996 establishes federal standards protecting sensitive health information from disclosure without patient's consent. The US Department of Health and Human Services issued the HIPAA Privacy Rule to implement HIPAA requirements. The HIPAA Security Rule protects specific information cover the Privacy Rule.”

Citation: US Department of Health and Human Services

HIPAA Security Rule Elements

1. Ensure the confidentiality, integrity, and availability of all ePHI they create, receive, maintain, or transmit.

2. Protect against reasonably anticipated threats to the security or integrity of the information.

3. Protect against reasonably anticipated, impermissible uses or disclosures.

4. Ensure compliance by their workforce.

Privacy Metric Tracking

93% tracked at least one privacy metric

14% used five or more

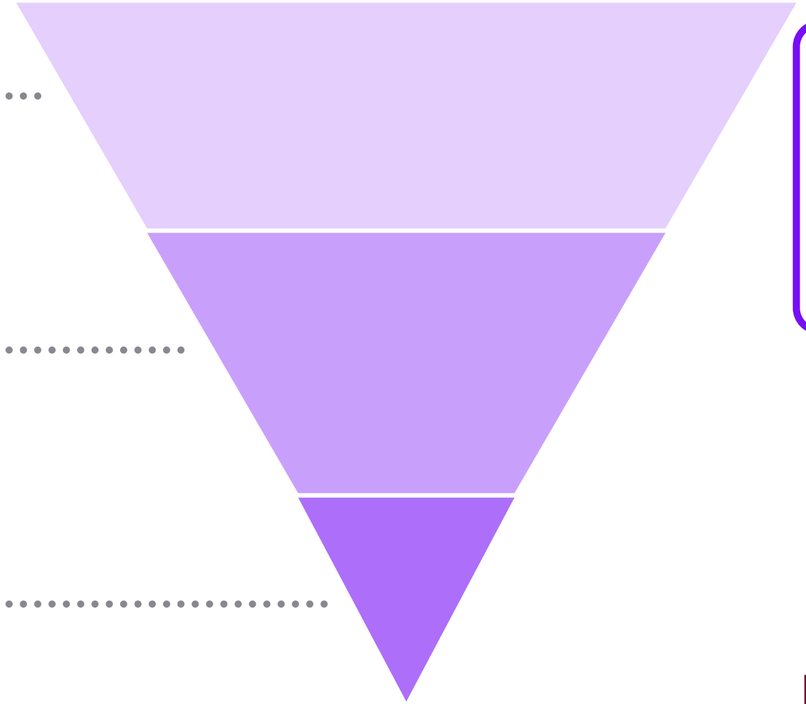
Privacy Metric Tracking

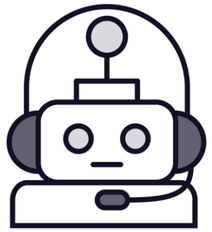
AI Privacy Actions

Internet Data Generated Daily
1,180,591,620,717,411,303,424 bits

Healthcare data generated daily
393,530,540,239,137,101,134 bits

Healthcare information effectively utilized daily,
11,790,000,000,000,000,000 bits





Privacy-Preserving Machine Learning (PPML)

- ❑ Differential privacy
- ❑ Secure multi-party computation
- ❑ Confidential computing
- ❑ Adversarial training
- ❑ Model watermarking
- ❑ Deliberate output manipulation

Privacy-preserving Machine Learning (PPML)

Differential privacy	Secure multi-party computation	Confidential computing
Adversarial training	Model watermarking	Deliberate output manipulation

Course Summary

Assess organizational alignment with the AI lifecycle

Review your most challenging areas of administrative overload

Apply safeguards to prevent harm

Thank you for your time!

@drlyronhandrews | www.profabula.com